# Hongyi Hao

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## Education

**Columbia University** — *New York, USA*
M.S. in Computer Engineering — *Sep. 2026 – Present*

**Shanxi Institute of Technology** — *Yangquan, China*
B.Eng. in Computer Science — *Sep. 2022 – Jul. 2026*

## Research Interests

**Core Research Interests:** My research lies at the intersection of reinforcement learning, multi-agent decision-making, and robot learning, with a focus on agents that act in complex, partially observed environments, from imperfect-information games to legged and mobile robots.

**Core Research Question:** How can we build learning-based agents that perceive, plan, and make reliable decisions in the real world, generalizing beyond the narrow tasks they were trained on?

## Honors and Awards (Selected)

2026 **Outstanding Graduate**, Shanxi Institute of Technology. Top 5% of department.
2025 **National Second Prize**, "Challenge Cup" National College Students' Extracurricular Academic and Scientific Works Competition.
2025 **Silver Award**, China International College Students' Innovation Competition.
2025 Outstanding Student, Shanxi Institute of Technology.
2024 Provincial Second Place, Shanxi Central Youth City Innovation and Entrepreneurship Competition.
2023 Third Prize, "Zhongkong Xinda Cup" Programming Contest, North China, Hong Kong, Macao & Taiwan.

## Publications

PREPRINTS

**[P2] Cross-Embodied Reinforcement Learning Control Method for Quadruped Robots on Complex Terrain.**
**Hongyi Hao**
*ICRSS*, 2025.

**[P1] Structured Pruning for Efficient YOLO11 Deployment: A Comparative Study of LAMP and SPPF.**
**Hongyi Hao**
*ICMAEE*, 2025.

JOURNAL ARTICLES

**[J2] EEGAuth: A Secure and Lightweight EEG-Based System Integrating Authentication and Key Generation.**
Xun Han, Jun Xiao, Yifan Liu, Ruilin Zhang, Kaibiao Zhu, **Hongyi Hao**, Youqi Li, Fan Li, Qian Zhang
*IEEE Internet of Things Journal*, 2025. [Paper]

**[J1] Data Security Strategies and Technologies for Robust Cloud Computing.**
**Hongyi Hao**
*Discover Applied Sciences*, 2025. [Paper]

CONFERENCE PROCEEDINGS

**[C7] Research and Application of Multi-Agent Cooperative Decision Algorithm Based on Deep Reinforcement Learning.**
**Hongyi Hao**
*ACESPE*, 2025. [Paper]

**[C6] Research on Multi-Robot Obstacle Avoidance Technology Based on HRVO Combined with Deep Reinforcement Learning.**
**Hongyi Hao**
*ICCASIT*, 2025. [Paper]
**[C5] Multidimensional Data Query Processing Methods Using Artificial Intelligence and Machine Learning Algorithms.**
**Hongyi Hao**
*SSITCON*, 2025. [Paper]
**[C4] VoteRFID: Multi-Target Rotation Recognition Algorithm Based on Phase Encoding via Markov Transition Field.**
Zhonghao Wang, **Hongyi Hao**, Biaokai Zhu, Yiran Li, Yaxuan Guo, Jiaqi He, Qian Zhang
*IEEE ISPA*, 2025. [Paper]
**[C3] Temperature-Aware Liquid Sensing with mm-Wave Radar.**
Zeyu Fan, Zhaokui Wang, **Hongyi Hao**, Zhao Lv, Ping Li
*IEEE ISPA*, 2025. [Paper]
**[C2] Stability Analysis of Polygonal Slopes Considering Spatial Variability of Soil Parameters.**
**Hongyi Hao**
*ICDA*, 2025. [Paper]
**[C1] ECHOES: Input Sensing and Reconstruction Model Utilizing CNN and LSTM with Mobile Sensor Data.**
**Hongyi Hao**
*IEEE ICICC*, 2025. [Paper]

# Research Experience

**Distributed Machine Learning** *Remote*
Online Research Seminar, supervised by Prof. Soummya Kar *May 2025 – Aug. 2025*

- Studied distributed and federated learning paradigms, covering supervised learning, stochastic optimization, distributed machine learning architectures, advanced federated learning, and convergence analysis.
- Implemented machine learning models and stochastic optimization methods on real-world datasets for image classification, house price prediction, and traffic monitoring.
- Contributed to the development of a communication-efficient federated learning variant for imbalanced datasets.
- Explored advanced applications of federated learning in autonomous systems and edge computing.

**SESAME Algorithm Club, Shanxi Institute of Technology** *Yangquan, China*
Independent Researcher *Sep. 2022 – Jul. 2026*

- Conducted independent project work in computer vision, deep reinforcement learning, multi-agent decision-making, object detection, wireless sensing, and acoustic sensing.
- Developed independent projects, with code available on GitHub.
- Participated in academic and programming competitions.

# Industry Experience

**Institute of Automation, Chinese Academy of Sciences** *Nanjing, China*
Backend Development Intern *Jul. 2024 – Sep. 2024*

- Performed basic backend development using Spring Boot.
- Studied MySQL and Oracle database principles, Spring Cloud microservices architecture, and distributed system workflows.

**ArcherMind Technology** *Nanjing, China*
Data Analyst Intern *Jun. 2023 – Aug. 2023*

- Collected and organized data generated by in-vehicle AI systems.
- Cleaned and filtered anomalous or unreasonable data to support the accuracy of AI-generated decisions and responses and to improve user experience.

## Academic Services

**Instructor** SESAME Algorithm Club: taught programming and algorithms to beginners *Sep. 2022 – Jul. 2026*

## Skills

**Programming** C/C++, Python, Java, SQL, Linux, React, MATLAB, LaTeX, Node.js
**Frameworks** PyTorch, TensorFlow, NumPy, OpenCV, ROS, Flask, Spring Boot, CMake, Git
**Hardware** Raspberry Pi, Arduino, ESP32

## Project Portfolio (Selected)

**MahJong_RLAI: Suphx-Style RL Agent for Riichi Mahjong** *Open Source*
Developer. [Code] *Nov. 2025 – Present*

- Implemented a reinforcement learning agent for Riichi Mahjong.
- Used the Tianshou reinforcement learning library for algorithm implementation.
- Designed the approach after Microsoft Research's Suphx Mahjong AI.

**Tunnel Concrete Defect Detection with Improved YOLO & Inspection Robot** *Shanxi, China*
Founder & Developer. [Introduction] *Sep. 2022 – Feb. 2026*

- Developed a YOLO11-based model to recognize defects in ground-penetrating radar waveforms of tunnel linings.
- Redesigned a negative-pressure adhesion robot for inspecting tunnel walls and ceilings.
- The project was applied in field inspections covering **68 tunnels** and more than **190 km** of survey lines.

- Built an improved YOLOv11 detector for identifying litter in cluttered roadside scenes.
- Continuously retrained the model on data collected during deployment.

**EchoEyeTracker: Acoustic Gaze Tracking** *Shanxi, China*
Developer. [Code] [Reference] *Mar. 2025 – Jul. 2026*

- Reproduced the GazeTrak acoustic eye-tracking pipeline.
- Improved pipeline robustness through modifications to training and data processing.